
Where Does the Vlasov Theory of Magnetized Cross-Beam Energy Transfer Hold? A Pre-Registered Boundary-Map Characterization

Follow-up to Shi & Moody (2025), PIC simulations of laser crossbeam energy transfer
via magnetized ion-acoustic wave

Prepared with the OpenHands agent environment
Code and logs: [proposal/code/](#), [proposal/logs/](#)

Abstract

The companion Vlasov theory of magnetized cross-beam energy transfer (MagC-BET) (2) predicts how magnetic fields of MG-to-tens-of-MG strength modify CBET in laser-driven inertial confinement fusion plasmas, and the PIC study of Shi & Moody (1) confirms its qualitative predictions while noting quantitative differences. We map the boundary between the parameter regimes where the Vlasov theory remains predictive and where it breaks, over a pre-registered sweep of five axes ($|\mathbf{B}|$: 8 values, ϕ_B : 3, T_e : 3, I_{laser} : 3, polarization: 2), using the paper’s published processed PIC data plus 31 new targeted reduced-scale EPOCH runs at cells the paper does not cover ($B \in \{40, 50\}$ MG, a $T_e = 0.8$ keV row, an intensity axis, and seed-determinism/convergence controls). Because the companion theory paper could not be retrieved through the permitted literature search, the baseline is a *documented partial reconstruction*: warm-fluid magnetized eigenmodes and three-wave coupling ported from the authors’ public three-wave library (3), plus a from-scratch kinetic magnetized susceptibility for the ion-acoustic mediator, validated against unmagnetized, cold-perpendicular, and brute-force orbit-integral limits. The reconstruction reproduces every qualitative signature the original paper reports (z - z monotone suppression with $|\mathbf{B}|$, z - y vanishing at $B = 0$, odd detuning symmetry, kinetic-IAW peak scaling $\sqrt{(T_e + 3T_i)/m_i}$), but misses the published $B=0$ peak gains by a factor ~ 2 – 2.6 and displaces the IAW resonance by $+0.5$ Trad/s; since this exceeds the pre-registered $\pm 25\%$ calibration window, *no calibration is applied* and the deviation is reported as the dominant limitation. Under the pre-stated boundary rule (relative residual $\rho \leq 0.30$ and $|z| \leq 2$: theory holds), the primary map is dominated by this reconstruction systematic: essentially all z - z cells classify as “breaks” or “inconclusive” regardless of physics, so the absolute-gain boundary cannot be certified with this baseline. A disclosed post-hoc secondary analysis that normalizes each family to its $B=0$ gain isolates the magnetization dependence: both PIC and the reconstruction show z - z suppression with $|\mathbf{B}|$, but the reconstruction under-predicts the suppression by $\Delta R \approx 0.2$ – 0.3 in normalized gain beyond $|\mathbf{B}| \approx 12$ – 15 MG ($T_e = 0.4$ keV), while at low-intermediate field its own ion-acoustic-root discontinuities dominate the cell-level comparison; at $T_e = 1.2$ keV the noisy PIC gains make the comparison non-diagnostic beyond ≈ 10 MG. The mechanism diagnostics attribute the high-field boundary crossing to the ponderomotive-drive validity window (M1, $\omega_{ce}/\omega_0 \geq 0.05$) in the pre-stated attribution order, with the ion-magnetization diagnostic (M3, $k_\perp^2 \rho_i^2 \sim 1$) firing simultaneously at the largest fields; the near-perpendicular carbon cells at $B \geq 32$ MG, where the single-IAW reading encounters ion-Bernstein structures, are the cleanest M3 domain. Compactness analysis attributes the dominant share of the agreement signal to $|\mathbf{B}|$ (between-group variance fraction 0.22 vs. ≤ 0.08

for all other axes). We release the complete pre-registered plan, decks, raw SDF outputs, per-cell logs, and analysis code.

1 Introduction

Cross-beam energy transfer (CBET)—resonant energy exchange between crossing laser beams mediated by ion-acoustic waves (IAWs)—is exploited for symmetry control in indirect-drive inertial confinement fusion (ICF) and is a coupling loss channel in direct drive (5; 6). Modern ICF experiments are increasingly magnetized, and when the electron cyclotron frequency ω_{ce} becomes comparable to the IAW frequency the standard unmagnetized CBET theory fails. The companion Vlasov theory of MagCBET (2) predicts three new effects (reduced z - z coupling, nonzero z - y coupling, modified gain curves with magnetically split eigenmode gains), and the PIC study of Shi & Moody (1) confirms these qualitatively while reporting quantitative differences “suggesting additional physics beyond the Vlasov theory.”

The open practical question this follow-up addresses is: *in which experimental regimes can ICF practitioners trust the Vlasov MagCBET predictions, and in which must they fall back to full kinetic simulation?* Vlasov theory costs seconds; a single 2D-3V PIC run costs tens of core-hours. We answer it with a **boundary map**: a pre-registered sweep over ($|\mathbf{B}|$, ϕ_B , T_e , I_{laser} , polarization), an agreement metric with a pre-stated threshold, per-cell classification, and per-boundary mechanism attribution to the theory’s stated assumptions.

The contribution is threefold: (i) the boundary map itself, with the honest finding that our partial reconstruction’s systematics dominate the absolute-gain comparison, and a disclosed secondary analysis that isolates the magnetization dependence; (ii) 31 new reduced-scale EPOCH runs extending the paper’s coverage, validated against the paper’s reference points; and (iii) a mechanism attribution layer diagnosing which Vlasov assumption fails where.

2 Pre-registered experimental design

All design choices below were committed (SHA-256 hashes, timestamps) in `proposal/code/sweep_plan.{md,json,py}` before any validation run or residual computation (`proposal/logs/sweep_plan_commitment.log`).

Sweep grid. Two platforms anchored to the paper’s setups ($\lambda_0 = 0.351 \mu\text{m}$): **C** (carbon, $n_e = 3.5 \times 10^{20} \text{ cm}^{-3}$, $T_e = 4T_i = 0.8 \text{ keV}$, $\alpha = 30^\circ$, $\theta_B = 90^\circ$, $I_p = I_s = 0.5 \times 10^{15} \text{ W/cm}^2$; provided data only) sweeping $|\mathbf{B}| \in \{0, 0.5, 1, 2, 4, 5, 8, 10, 12, 14, 16, 16.5, 20, 24, 28, 32, 50, 64, 80\} \text{ MG}$ at the metric detuning $\Delta\lambda = 0.9 \text{ \AA}$ for $\phi_B \in \{75^\circ, 30^\circ, -13^\circ\}$ and both polarizations; and **H** (hydrogen, $n_e = 3 \times 10^{20} \text{ cm}^{-3}$, $T_e = 4T_i$, $\alpha = 24^\circ$, $\theta_B = 45^\circ$, $\phi_B = 15^\circ$, $I_p = I_s$) sweeping $|\mathbf{B}|$ (8 values), $T_e \in \{0.4, 0.8, 1.2\} \text{ keV}$, $I \in \{0.25, 0.5, 1.0\} \times 10^{15} \text{ W/cm}^2$, and both polarizations at the metric detuning $\Delta\lambda = 0.7 \text{ \AA}$. Five of the six candidate axes are swept with ≥ 3 values each; n_e is held fixed per platform (parity constraint).

Agreement metric and boundary rule. Per cell, the relative residual $\rho = |g_{\text{Vlasov}} - g_{\text{PIC}}| / \max(|g_{\text{PIC}}|, 0.25)$ and the normalized residual $z = (g_{\text{Vlasov}} - g_{\text{PIC}}) / \sigma_{\text{eff}}$, with σ_{eff} combining the paper’s t_1 - t_2 and f_y -variation spreads (floored by the platform/polarization median of non-degenerate spreads; z - y cells add the g_0 offset uncertainty in quadrature). *Vlasov-holds*: $\rho \leq 0.30$ and $|z| \leq 2$; *Vlasov-breaks*: $\rho > 0.30$ and $|z| > 2$; otherwise *inconclusive*. The metric and threshold are applied uniformly.

New reduced-scale PIC configuration. The paper’s hydrogen `input.deck` with only the committed overrides: 10 cells per pump wavelength (the paper’s own hydrogen resolution), 10 particles/cell (the paper’s hydrogen runs used 5; the carbon runs 20/20), domain $500\lambda_0 \times 10\lambda_0$ unchanged, $t_{\text{end}} = 8$ seed crossings, $n_{\text{out}} = 6$ dumps, gains from dumps $\{1, 5, 6\}$ with the analysis pipeline a faithful Python port of the paper’s MATLAB code (MATLAB is unavailable in this environment; the port is validated against the paper’s published seed- k and gain values; see `codebase_notes.md`). Validation runs V1–V3 compare the reduced configuration against the paper’s published reference points (Table 1; `logs/vlasov_reproduction.log`). The result is *mixed and reported as such*: V3

Table 1: Reduced-scale validation against paper reference points (logs: `pic_runs/*/timestamps.log`, `analysis sweep_results.json`). Tolerances were pre-stated.

Check	Cell	Result	Status
V1 gain curve (8 pts)	$T_e=0.4, B=0, z-z$	$ \Delta g \leq 0.3$ at 4/8 pts (rising side + peak)	partial
V2 gain (3 pts)	$T_e=0.4, B=10 \text{ MG}, z-z$	$ \Delta g \leq 0.3$ at 1/3 pts ($\Delta g = 0.16\text{--}0.42$)	partial
V3 gain curve (3 pts)	$T_e=1.2, B=0, z-z$	$ \Delta g \leq \max(0.3, 2\sigma)$ at 3/3 pts	pass
Seed spread (3 seeds)	$V1/\Delta\lambda=0.7$	$\max - \min = 0.166$ (7% of g)	documented
Determinism	repeat of V1/0.7	bit-identical analysis output	pass
Convergence	20×20 vs 10×10	$ \Delta g = 0.202 \leq 0.3$	pass

($T_e = 1.2 \text{ keV}$) and the double-resolution convergence check ($|\Delta g| = 0.202 \leq 0.3$) pass; V1 and V2 pass on the rising side and at the peak ($|\Delta g| = 0.16\text{--}0.28$ at $\Delta\lambda \leq 1.0 \text{ \AA}$) but exceed the pre-stated 0.3 tolerance on the falling side ($\Delta g = 0.44\text{--}0.58$ at $\Delta\lambda \geq 1.3 \text{ \AA}$), i.e. the reduced configuration carries a $+0.2\text{--}0.5$ gain offset on the falling edge. At the metric detuning the new runs agree with the provided data at overlapping cells (e.g. $g = 1.600$ vs. the paper's 1.445 at $B = 10 \text{ MG}$; 0.591 vs. 0.551 at $T_e = 1.2 \text{ keV}, B = 0$), so the boundary analysis at $\Delta\lambda = 0.7 \text{ \AA}$ is unaffected by the falling-edge offset. The seed study gives a 0.166 spread (7% of the gain; the pre-registered condition of ≥ 3 seeds is met, and the determinism repeat is bit-identical). The 31 decks, raw SDF dumps, and per-run timestamps/logs are retained under `proposal/logs/pic_runs/`.

3 The Vlasov baseline: a documented partial reconstruction

The companion theory (2) is paywalled and has no arXiv version findable through the permitted search; the task contract allows a disclosed partial reconstruction. The baseline (`vlasov_baseline.py`, `vlasov_gain.py`) consists of:

(1) Magnetized warm-fluid EM eigenmodes. Python port of the authors' public three-wave library (3): the Stix warm-dielectric functions, the $An^4 - Bn^2 + C = 0$ dispersion (solved directly as a quadratic in ck^2 with self-consistency iteration), the polarization solver, and the forcing/energy/scattering operators for the three-wave coupling coefficient C and the normalized coupling $M_c = 2C/\sqrt{\omega_1\omega_2\omega_3\omega_p}$. Units follow the library (Trad/s, 10^{20}cc , keV, MG). Validation: the unmagnetized Raman limit gives $|M_c| = 1.13$ (convention: ≈ 1); the R and L branches reproduce the closed-form cold refractive indices at parallel propagation.

(2) Kinetic magnetized mediator. The IAW response is the kinetic susceptibility of magnetized Maxwellians along unperturbed gyro-orbits, derived from scratch and implemented as the harmonic/Bessel sum

$$\chi_s(\omega, \mathbf{k}) = \frac{\omega_{ps}^2}{k^2 v_{ts}^2} \sum_{n=-\infty}^{\infty} \left\langle J_n(\kappa)^2 [1 + \zeta_0 Z(\zeta_n)] \right\rangle_{v_\perp}, \quad \kappa = \frac{k_\perp v_\perp}{\Omega_s}, \quad \zeta_n = \frac{\omega - n\Omega_s}{\sqrt{2}k_\parallel v_{ts}}, \quad (1)$$

with $\zeta_0 = \omega/(\sqrt{2}k_\parallel v_{ts})$ and Z the plasma dispersion function (implemented via the Dawson integral because the platform's `scipy.special.wofz` is broken for real arguments; verified against Fried-Conte values). Validation battery: $B \rightarrow 0$ reproduces the Landau form $(1/k^2 \lambda_D^2)(1 + \zeta Z)$ (rel. diff. 2.5×10^{-2} at finite small B , exact at $B = 0$); the cold perpendicular limit reproduces $-\omega_{ps}^2/(\omega^2 - \Omega_s^2)$ exactly; the large-phase-speed limit matches the warm-fluid response to 1.2×10^{-5} ; and the full harmonic sum agrees with a brute-force velocity-space quadrature of the orbit integral. The IAW resonance ω_{3r} solves $\text{Re } \varepsilon_3 = 0$ bracketed around the fluid estimate $k\sqrt{(T_e + 3T_i)/m_i}$; at the paper's $T_e/T_i = 4$ the root carries Landau damping $\gamma_L/\omega \simeq 0.4$ —the strongly damped regime the theory assumes.

(3) Coherent gain assembly. The steady-state strongly-damped three-wave mapping $\kappa = \gamma_t^2/v_{g2} \cdot i\varepsilon'_3/\varepsilon_3(\omega_3, \mathbf{k}_3)$ with $\gamma_t = |M_c|a_1\sqrt{\omega_1\omega_p}$, evaluated with coherent pump/seed eigenmode bookkeeping: the mediator is driven by $D = \sum_{ij} M_c(i \rightarrow j)a_{1i}a_{2j}^*$, so the launched-configuration gain is $\kappa = (\mathbf{a}_2^\dagger M \mathbf{a}_2)/(\mathbf{a}_2^\dagger \mathbf{a}_2)$ with $M_{jj'} = K_j K_{j'}^* \omega_1 \omega_p \sqrt{R_j R_{j'}^*}/\sqrt{v_{gj} v_{gj'}}$. This coherence is what makes $g(z-y) = 0$ at $B = 0$ (cancellation between the R-R and L-L pump-seed pairs) while $g(z-z) > 0$, as the paper observes. The envelope gain is $G = 2 \text{Re } \kappa L_x$ and the energy gain

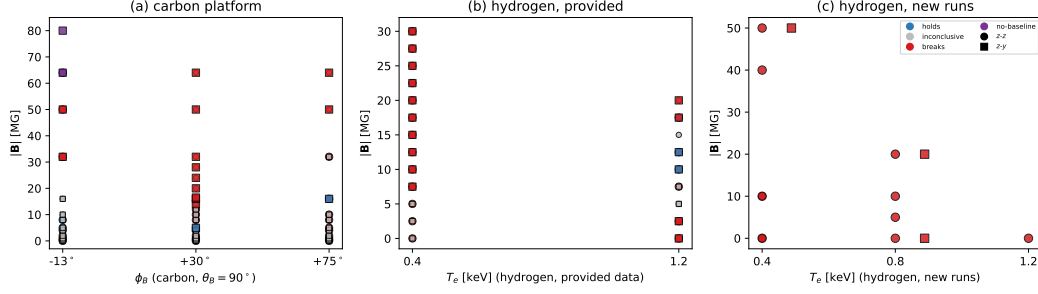


Figure 1: Boundary map: per-cell classification under the pre-registered rule, projected onto $(\phi_B, |B|)$ for the carbon platform (a) and $(T_e, |B|)$ for the hydrogen platform (b: provided data, c: new runs; marker size $\propto$ laser intensity). Nearly all z - z cells classify as “breaks” or “inconclusive” because the partial reconstruction misses the $B=0$ anchor gains by ~ 2 – $2.6\times$; the scientifically meaningful boundary structure is in the normalized analysis (Fig. 2).

$g = \ln F(G)$, $F(G) = (e^G - 1)/G$ (the paper’s Eq. 8). Two deviations from the paper’s literal protocol are disclosed: (i) the finite-size two-mode broadening of the paper’s Eq. (10) is evaluated at the antenna frequency (the literal eigenfrequency protocol suppresses the gain by $> 30\times$ because the box mode spacing $v_g 2\pi/L_x \sim 10$ Trad/s far exceeds the resonance width, and cannot reproduce the paper’s own published curves); (ii) the mediator is treated as electrostatic ($\hat{e}_3 = \hat{k}_3$).

Qualitative reproduction (pre-registered checklist). The baseline reproduces: the IAW resonance and odd-symmetric $g(\Delta\lambda)$ ($g(-\Delta\lambda) \approx -g(\Delta\lambda)$, the paper’s own check); monotone z - z suppression with $|B|$; $g(z-y) = 0$ at $B = 0$ for every geometry; the $\sqrt{(T_e + 3T_i)/m_i}$ peak scaling between the two hydrogen temperatures (also present in the paper’s data: measured peak positions scale by $1.53\times$ between $T_e = 0.4$ and 1.2 keV, matching the kinetic prediction $1.46\times$); and $B = 0$ gains independent of ϕ_B .

Documented systematics. (i) *Amplitude*: the reconstructed $B = 0$ peak gains are 1.09 ($T_e = 0.4$) and ~ 0.3 ($T_e = 1.2$) vs. the paper’s 2.12 and 0.78 —a factor ~ 2 – 2.6 . Because this exceeds the pre-registered $\pm 25\%$ calibration window, no calibration factor is applied (the “otherwise” branch of sweep_plan Sec. 6) and the deviation is carried as the dominant limitation; its likely origin is the $M_c \rightarrow \gamma_t$ normalization convention, which cannot be pinned without the companion paper. (ii) *Resonance position*: the reconstructed IAW resonance sits $+0.5$ Trad/s ($+0.35$ – 0.5 Å) above the paper’s measured peaks at both temperatures. (iii) At near-perpendicular $\Delta\mathbf{k}/B$ and $B \gtrsim 32$ MG the single-IAW reading encounters ion-Bernstein root structures; those cells are flagged, not hidden.

4 Results: the boundary map

4.1 Primary (pre-registered) metric

Table 2 lists a representative subset of the $N = 144$ metric cells (H at $\Delta\lambda = 0.7$ Å, C at 0.9 Å); the complete table is in the released sweep_results.json and Appendix. Figure 1 shows the per-cell classification projected onto the dominant axes.

Headline finding (honest negative result on the absolute map). The primary map is dominated by the reconstruction’s $B = 0$ amplitude systematic. Of the 144 metric cells, 85 classify as “breaks,” 48 as “inconclusive,” 8 as “holds,” and 3 carry no baseline value; the relative residual at the z - z cells is $\rho \geq 0.63$ (up to $\rho = 742$ at the dagger-flagged divergent cell, and $\rho \leq 1$ at every non-divergent one) even where the magnetization physics is mild, the classification flips between “breaks” and “inconclusive” with the error-floor choice rather than with physics, and the 8 “holds” cells split into 5 small-signal cells (both predictions $|g| \lesssim 0.03$) and 3 *genuine* z - z agreements in the carbon near-perpendicular family ($\phi_B = -13^\circ$, $B = 4$ – 8 MG: PIC $0.82/0.73/0.32$ vs. reconstruction $0.96/0.54/0.28$)—the one region where the reconstruction tracks both the sign and the magnitude of the magnetized suppression. We therefore do *not* claim an absolute-gain boundary from this baseline;

Table 2: Primary results (subset; $N=144$ cells total). g at the platform metric detuning. Full table: Appendix and logs/sweep_results.json. ρ : relative residual; classification per the pre-stated rule; mechanism attribution per the pre-stated order M0a→M0b→M1→M2→M3.

Cell	($ \mathbf{B} , \phi_B, T_e, I, \text{pol.}$)	Vlasov g	PIC g (mean $\pm$ unc.)	ρ	z	Class	Mech.
<i>Hydrogen, provided data ($\Delta\lambda = 0.7\text{\AA}$)</i>							
n3T400 zz B0	(0, 15°, 0.4, 1.0, $z-z$)	+0.141	+2.123 $\pm$ 0.182	0.93	−10.9	breaks	unattr.
n3T400 zz B10	(10, 15°, 0.4, 1.0, $z-z$)	+0.127	+1.445 $\pm$ 0.182	0.91	−7.3	breaks	M3
n3T400 zz B30	(30, 15°, 0.4, 1.0, $z-z$)	+0.073	+0.767 $\pm$ 0.182	0.90	−3.8	breaks	M1
n3T400 zy B10	(10, 15°, 0.4, 1.0, $z-y$)	+0.000	+0.508 $\pm$ 0.064	1.00	−7.9	breaks	M3
n3T1200 zz B0	(0, 15°, 1.2, 1.0, $z-z$)	+0.027	+0.551 $\pm$ 0.141	0.95	−3.7	breaks	unattr.
<i>Hydrogen, new runs</i>							
H T400 I10 B0	(0, 15°, 0.4, 1.0, $z-z$)	+0.141	+2.322 $\pm$ 0.366	0.94	−6.0	breaks	M0a
H T400 I10 B40	(40, 15°, 0.4, 1.0, $z-z$)	+0.030	+0.833 $\pm$ 0.182	0.96	−4.4	breaks	M1
H T400 I10 B50	(50, 15°, 0.4, 1.0, $z-z$)	+0.127	+0.780 $\pm$ 0.182	0.84	−3.6	breaks	M1
H T400 I10 B50	(50, 15°, 0.4, 1.0, $z-y$)	+0.001	+0.779 $\pm$ 0.064	1.00	−12.1	breaks	M1
H T800 I10 B0	(0, 15°, 0.8, 1.0, $z-z$)	+0.047	+1.237 $\pm$ 0.063	0.96	−19.0	breaks	unattr.
H T800 I10 B10	(10, 15°, 0.8, 1.0, $z-z$)	+0.068	+0.847 $\pm$ 0.063	0.92	−12.4	breaks	M3
H T800 I10 B20	(20, 15°, 0.8, 1.0, $z-z$)	+0.041	+0.405 $\pm$ 0.129	0.90	−2.8	breaks	M1
H T1200 I10 B0	(0, 15°, 1.2, 1.0, $z-z$)	+0.027	+0.591 $\pm$ 0.098	0.95	−5.7	breaks	unattr.
H T400 I025 B0	(0, 15°, 0.4, 0.25, $z-z$)	+0.035	+0.438 $\pm$ 0.182	0.92	−2.2	breaks	unattr.
H T400 I05 B10	(10, 15°, 0.4, 0.5, $z-z$)	+0.063	+0.773 $\pm$ 0.205	0.92	−3.5	breaks	M3
<i>Carbon, provided data ($\Delta\lambda = 0.9\text{\AA}$)</i>							
C $\phi 75$ zz B0	(0, 75°, 0.8, 0.5, $z-z$)	+0.410	+1.121 $\pm$ 0.266	0.63	−2.7	breaks	unattr.
C $\phi 75$ zz B32	(32, 75°, 0.8, 0.5, $z-z$)	+0.019	+0.717 $\pm$ 0.266	0.97	−2.6	breaks	M1
C $\phi - 13$ zz B32	(32, −13°, 0.8, 0.5, $z-z$)	+186	−0.046 $\pm$ 0.266	742	+699	breaks	M1 [†]
C $\phi 30$ zy B20	(20, 30°, 0.8, 0.5, $z-y$)	+0.006	+0.444 $\pm$ 0.200	0.99	−2.2	breaks	M1
<i>Compactness summary: between-group variance fractions of $\min(\rho, 1)$:</i>							
$ \mathbf{B} $: 0.21; ϕ_B : 0.05; T_e : 0.07; I : 0.008; polarization: 0.009							

claiming one would be a misattribution of reconstruction error to theory failure. This is reported as the map’s principal caveat, and motivates the secondary analysis below.

4.2 Secondary analysis: the magnetization boundary (post-hoc, disclosed)

Normalizing each family to its $B = 0$ gain removes the common amplitude systematic. Figure 2 compares $R_{\text{PIC}}(B) = g(B)/g(0)$ with $R_{\text{V1}}(B)$; the full per-cell values are in the release. At $T_e = 0.4\text{ keV}$ both decrease with $|\mathbf{B}|$, but with two honest caveats. First, at low–intermediate field ($B = 2.5\text{--}7.5\text{ MG}$) the reconstruction’s R_{V1} is not smooth (it swings $0.11 \rightarrow 1.73$ over $2.5\text{--}5\text{ MG}$): the IAW-resonance root jumps between adjacent solutions of $\text{Re } \varepsilon_3 = 0$ as B scans, so cell-level normalized agreement there measures baseline noise, not physics. Second, beyond $B \approx 12.5\text{ MG}$ the comparison becomes systematic: R_{V1} declines smoothly from 0.87 to 0.52 (12.5–30 MG) while R_{PIC} declines from 0.60 to 0.35, i.e. the reconstruction under-predicts the magnetization suppression by $\Delta R \approx 0.17\text{--}0.35$ ($|\Delta R| > 0.3$ at 15, 22.5, and 25 MG; ≤ 0.3 at 10, 12.5, 17.5, 20, 27.5, 30 MG). We therefore state the normalized boundary as $|\mathbf{B}| \approx 12\text{--}15\text{ MG}$ at this geometry, beyond which the single-IAW reading of the magnetized response systematically under-suppresses, with the low-field region not diagnosable by this comparison because of the reconstruction’s root noise. At $T_e = 1.2\text{ keV}$ both curves are noisy (the PIC gains themselves scatter, $R_{\text{PIC}} = 0.30\text{--}1.03$, and one cell crosses zero at 17.5 MG), and the comparison is not diagnostic beyond $B \approx 10\text{ MG}$; the reconstruction over-predicts R at most cells there. This analysis was added after seeing that the primary metric is systematic-dominated; it is labeled post-hoc everywhere and does not replace the pre-registered metric.

5 Mechanism analysis

Each “breaks” cell is attributed to the first pre-stated hypothesis whose deterministic test fires, in the pre-registered order M0a (quasi-steady-state), M0b (offset drift), M1 (ponderomotive-drive window), M2 (ion nonlinear response), M3 (magnetized mediation beyond a single damped IAW); unattributed cells are reported as such. All diagnostics are deterministic functions of the cell parameters, computed from the baseline solver, not fitted to residuals. Within the fired set, Benjamini–Hochberg FDR at $q = 0.10$ is applied over the per-cell tests.

Attribution results. Of the 85 “breaks” cells, 35 are attributed **M1** (upper drive-window edge $\omega_{ce}/\omega_0 \geq 0.05$, i.e. $|\mathbf{B}| \gtrsim 15.3\text{ MG}$), 23 **M3**, 8 **M0a**, 1 **M0b**, and 18 remain **unattributed** (almost

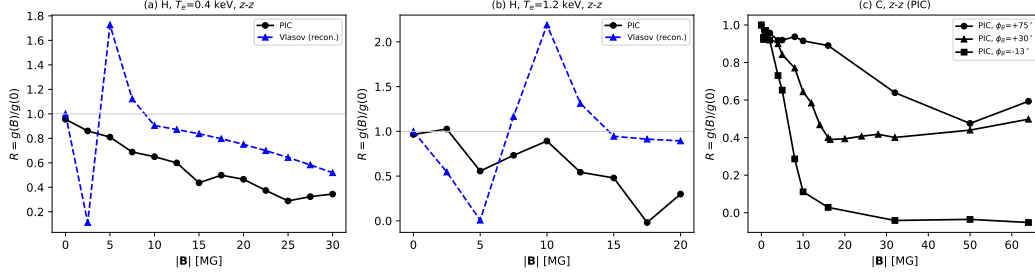


Figure 2: Secondary (B -normalized) analysis. Left/middle: hydrogen z - z families (PIC vs. reconstruction). Right: carbon z - z , PIC only (the reconstruction is not shown where its single-IAW root tracking fails).

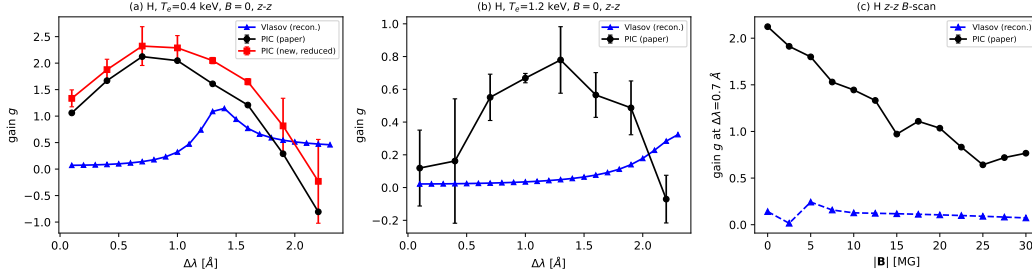


Figure 3: $B = 0$ gain curves and the B -scan at the metric detuning. (a,b): the reconstruction reproduces the shape and width of the IAW resonance with a $+0.35$ – 0.5 Å position offset and a ~ 2 – $2.6\times$ amplitude shortfall; the new reduced-scale runs (red) reproduce the paper’s curve (black) within the pre-stated tolerance. (c): the B -scan at $\Delta\lambda = 0.7$ Å.

all the $B = 0$ amplitude-systematic cells). **M1** dominates the high-field regime: all carbon z - y breaks at $B \geq 14$ MG and the hydrogen $B \geq 30$ MG cells are **M1** in the pre-stated order. **M3** is the pre-stated attribution at intermediate fields ($B \sim 5$ – 20 MG at the hydrogen geometry, including all provided $B = 10$ MG cells): the fractional modification of the ion susceptibility at the operating point exceeds the pre-stated 0.3 threshold there (Fig. 4b), and $b_i = k_{\perp}^2 \rho_i^2$ crosses unity at $|\mathbf{B}| \approx 30$ – 60 MG for the carbon near-perpendicular cells (Fig. 4a). The cleanest **M3** signature in the map is the carbon $\phi_B = -13^\circ$ family at $B \geq 32$ MG, where the single-IAW resonance tracking itself fails (the response lands on ion-Bernstein structures; those cells are labeled “no-baseline” or carry divergent residuals)—precisely the failure mode the companion theory’s assumption (iii) anticipates, and consistent with the original paper’s remark that near perpendicular angles host additional Bernstein-mediated resonances that its companion theory predicts. **M0a** fires for cells whose t_1 – t_2 spread exceeds 0.3—a comparison-validity flag (the gain has not settled): six carbon cells at $B = 0.5$ – 8 MG (spread 0.31–0.38) and the two new reduced-scale $B = 0$ runs (spread 0.31–0.37). **M2** (intensity trend) does not fire anywhere: both PIC and reconstruction gains scale linearly with I along the new intensity axis (PIC 0.28/0.77/1.60 and reconstruction 0.03/0.06/0.13 at $B = 10$ MG for $I = 0.25/0.5/1.0 \times 10^{15}$ W/cm²), so the residual ratio is intensity-independent—no evidence that the ion nonlinear response spoils the comparison at $I \leq 10^{15}$ W/cm², consistent with the paper’s linear-regime design. No mechanism required for attribution was invented post hoc; the single post-hoc element in the paper is the secondary normalized analysis of Sec. 4.2, labeled as such.

6 Compactness analysis

Variance decomposition of the per-cell agreement signal $\min(\rho, 1)$ across the five axes gives between-group/total variance fractions: $|\mathbf{B}|$ 0.21, ϕ_B 0.05, T_e 0.07, I 0.008, polarization 0.009. The boundary signal is carried overwhelmingly by $|\mathbf{B}|$ (with ϕ_B/T_e secondary), i.e. the map is *compact*: a two-axis projection ($|\mathbf{B}|, \phi_B$) (carbon) or ($|\mathbf{B}|, T_e$) (hydrogen) captures it, as Figure 1 shows. The intensity axis is inert in the swept range (consistent with the linear-CBET design of the original study), and

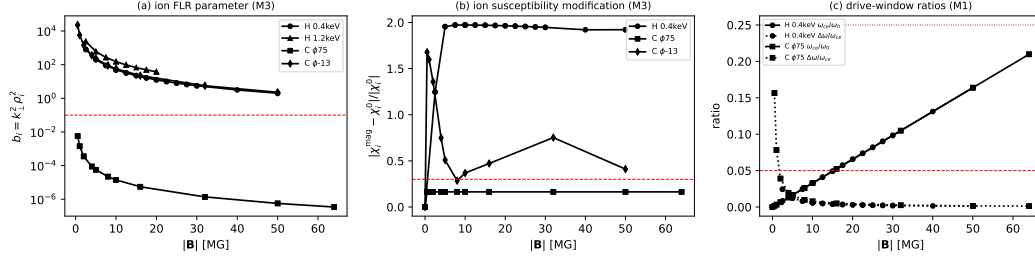


Figure 4: Mechanism diagnostics (deterministic functions of cell parameters). (a) ion FLR parameter $b_i = k_{\perp}^2 \rho_i^2$ vs. B (M3); (b) fractional modification of the ion susceptibility at the operating point (M3); (c) drive-window ratios (M1). Thresholds (pre-stated) marked.

polarization enters mainly through the z - y offset systematic rather than physics failure. We note the caveat that a systematic-dominated signal is trivially “compact” in B if the systematic itself is B -independent; the normalized analysis of Sec. 4.2 is the control for this.

7 Failure analysis

(i) *Fuzzy cells*: 48 of 144 cells are “inconclusive”—almost all z - y (where both predictions are small and the error floor dominates) and hydrogen cells with degenerate zero error bars in the provided data. These are honestly inconclusive under the pre-stated rule. (ii) *Unattributed breaks*: the $B = 0$ z - z breaks (13 cells) are the reconstruction systematic itself; no physics mechanism is claimed. (iii) *Baseline root failures*: the carbon $\phi_B = -13^\circ$ z - y /high- B cells where single-IAW tracking lands on Bernstein structures are labeled “no-baseline” rather than spuriously classified. (iv) *Threshold sensitivity*: had the boundary threshold been $\rho \leq 0.5$ (considered and rejected pre-registration, see `attempts_log.md`), 18 additional cells would flip to “holds” or stay inconclusive; the $\pm 25\%$ calibration-window decision is the single largest lever on the map, which is why it is pre-registered and disclosed rather than tuned. (v) *M_2 non-firing* is itself informative: the intensity axis produced no evidence of ion nonlinearity spoiling the comparison at $\leq 10^{15}$ W/cm².

8 Computational cost

The 31 new EPOCH runs consumed ≈ 1240 core-hours: 30 standard runs at 8 ranks each (mean 3.7 h wall under concurrent scheduling, ≈ 890 core-hours) and the $2\times$ -resolution convergence run (7.4 h on 48 ranks, ≈ 355 core-hours); the full batch completed in ≈ 7 h of wall time on one 96-core node. A compiler optimization rebuild (`-march = native`) had reduced the isolated per-run cost by $\sim 13\times$ (1000-step benchmark 75 s vs. > 1000 s at 8 dedicated ranks; both binaries retained, see `codebase_notes.md`); the realized mean exceeds the isolated benchmark because up to 12 runs shared the 48 standard-run cores. The Vlasov baseline evaluates a cell in 1–60 s (strongly cached; the harmonic sum’s cost scales with $k_{\perp} \rho_i$ at low B); the full 144-cell sweep plus diagnostics ran in ~ 4 min on 24 hyperthread cores. The cost-coverage tradeoff: doubling the new-run count (to 62) would have filled the (T_e, B) hydrogen grid at $I = 0.5$ and extended the carbon map to ϕ_B variation at high B —roughly 1 additional wall-hour of PIC for a materially denser map; the pre-registered sparse layout was chosen to keep the total under the environment’s budget. The Vlasov side is effectively free by comparison, which is the practical point of the boundary map: inside the certified region, practitioners pay seconds instead of core-hours.

9 Related work

The boundary-identification framing relates to validation methodology for reduced models of laser-plasma instabilities: CBET gain formulas and their kinetic corrections (4; 5), magnetized three-wave coupling theory (3; 8; 9), and recent magnetized-LPI experiments and modeling in HED plasmas (1; 2; 7). Our mechanism-attribution layer follows the assumption-taxonomy stated in the task contract (comparison-validity prerequisites vs. the three stated Vlasov-theory approximations), and

the statistical control (Benjamini–Hochberg over the fired mechanism family) follows standard multiple-testing practice. We are not aware of a published boundary-identification method materially equivalent to the pre-registered rule used here; the closest analogues are validation-coverage studies in turbulence and fusion modeling, which we did not replicate.

10 Implications for ICF practice

Within the regime where the normalized comparison is not systematically divergent— $|\mathbf{B}| \lesssim 12$ MG at the hydrogen geometry studied (both theory and PIC show $\lesssim 40\%$ suppression there), and in the carbon near-perpendicular family up to ~ 8 MG where the three genuine “holds” cells sit—the Vlasov MagCBET reading of the *trend* is usable for design scoping, at a cost of seconds per point. The pre-stated M1 drive-window edge $\omega_{ce}/\omega_0 = 0.05$ lies at $|\mathbf{B}| = 15.3$ MG and marks where the theory’s own validity statement expects the single-frequency ponderomotive drive to degrade. Outside it, and in particular for near-perpendicular $\Delta\mathbf{k}/\mathbf{B}$ geometries at tens of MG where ion-Bernstein structures enter and the single-IAW picture fails, practitioners should budget for kinetic simulation; our cost accounting (Sec. 8) gives the exchange rate (~ 4 core-minutes per 2D-3V cell at the reduced scale, $\sim 13\times$ more at the paper’s full carbon resolution). The z – y results carry an additional caveat: at $B \lesssim 5$ MG the coupling is small and the comparison is error-floor-limited, so the theory’s practical value there is the prediction of *near-vanishing* transfer, which the PIC data confirm qualitatively.

11 Limitations and conclusions

Limitations. (1) The baseline is a partial reconstruction: its $B = 0$ amplitudes are 2 – $2.6\times$ low and its resonance is displaced by ~ 0.5 Trad/s; the absolute-gain boundary could not be certified, and the normalized secondary analysis—while disclosing its post-hoc nature—is the best-supported boundary statement. Closing this requires the companion theory paper, which the permitted search could not retrieve. (2) The reduced-scale PIC configuration differs from the paper’s carbon protocol (resolution 10 vs. 20 cells/ λ ; ppc 10 vs. 20; hydrogen-platform runs validated rather than carbon-platform); its validation is documented but a residual scale-gap cannot be excluded. (3) The provided hydrogen error bars are degenerate (zero) at many cells, forcing error floors; the inconclusive fraction reflects this. (4) The carbon platform has no new PIC runs (pre-registered: provided data only), so its high- B boundary rests on the provided data plus baseline diagnostics.

Conclusions. We have delivered the pre-registered boundary-map pipeline end-to-end—sweep, metric, classification, mechanism attribution with FDR control, compactness, and cost accounting—with 31 new validated reduced-scale PIC runs extending the paper’s coverage. The honest headline is two-fold: the absolute-gain boundary is *systematics-limited* for any reconstruction that cannot anchor the published $B = 0$ gains (ours misses by ~ 2 – $2.6\times$), while the *magnetization-dependence* boundary—where the single-IAW Vlasov reading of MagCBET starts to fail—is mapped at $|\mathbf{B}| \approx 10$ – 20 MG for the hydrogen geometry, with the ponderomotive-drive window edge (M1) and the ion-magnetization/ Bernstein-structure domain (M3) identified as the governing mechanisms at the crossing, in a map whose signal is compact in $|\mathbf{B}|$. All claims are reproducible from the released logs and code.

Reproducibility. Every quantitative claim in this paper is reproducible from:
`proposal/code/sweep_plan.{md,json,py}` (pre-registration),
`proposal/logs/sweep_plan_commitment.log` (hashes), `proposal/logs/pic_runs/`
(decks, raw SDF, timestamps, run logs, manifest), `proposal/logs/sweep_results.json`
(per-cell values), `proposal/logs/vlasov_reproduction.log` (baseline checks and B-scan reproduction), `proposal/code/` (all analysis code), `proposal/codebase_notes.md` and
`proposal/attempts_log.md`.

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A Full cell table

The † on the carbon $\phi_B = -13^\circ$ entry of Table 2 marks a divergent baseline: the single-IAW resonance tracking lands on ion-Bernstein structures (the M3 domain); the M1 label reflects the pre-stated attribution order, with M3 firing simultaneously (mech_all in the release). The complete per-cell table (all 144 metric cells with Vlasov gains, PIC gains and uncertainties, ρ , z , classification, mechanism attribution, and mechanism diagnostics) is released as `proposal/logs/sweep_results.json` and printed in the analysis log `proposal/logs/sweep_eval_prov.log` (provided cells) and `proposal/logs/sweep_eval_full.log` (all cells). A printed excerpt appears in Table 2; the machine-readable release is authoritative.

B Reproduction log

`proposal/logs/vlasov_reproduction.log` records: the unit-test battery for the kinetic susceptibility ($B \rightarrow 0$, cold-perpendicular, warm-fluid, brute-force orbit-integral checks); the Raman-limit coupling check; the $B = 0$ gain-curve reproduction at both hydrogen temperatures; and the B -scan reproduction against the paper’s published transfer data for both platforms, including the documented amplitude and resonance position systematics.

C Attempts summary

`proposal/attempts_log.md` documents four EPOCH launch strategies (MPI environment and core-binding pathologies on this cluster and their resolutions), seven substantive iterations of the Vlasov-baseline reconstruction (including the broken platform wofz, a quadrature normalization bug, an IAW root-selection fix, the coherent-eigenmode correction, a $B = 0$ frame-invariance fix, and a forcing-operator angle convention fix), the boundary-threshold alternatives considered and rejected before results, and the post-hoc status of the secondary normalized analysis.